MAYANK AGRAWAL

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EDUCATION

Bachelors Of Technology, Mechanical Eng., Indian Institute of Technology — July 2015 - May 2019
Ropar, Punjab

SKILLS

Technical Skills Ruby/Rails, Python, JavaScript/TypeScript, React, Scala, SQL, PostgreSQL, MySQL, MongoDB, Redis, Kafka, Sidekiq, AWS, Docker, Kubernetes, Nginx, MVC, REST APIs, background jobs, NETCONF/YANG, XPath, IOS-XE, TDL/Quark, LLM/RAG-based internal tooling

EXPERIENCE

Senior Software Engineer — June 2024 - Present
Cisco Systems / Cisco Meraki — *Bengaluru, Karnataka*

- Catalyst Dashboard, Simplified Onboarding & Config Analyzer
  - Owned major parts of Cisco Meraki Catalyst Simplified Onboarding and Config Analyzer, converting raw IOS-XE configs into a migration-readiness workflow.
  - Built XPath/CLI mapping, regex generation, config sanitization, CLI normalization, duplicate removal, and YANG/XPath automation flows.
  - Implemented asynchronous Config Analyzer API processing with background job polling for large customer configuration files.
  - Delivered supported, needs-review, ignored/system-generated, and equivalent-supported classification logic to reduce false migration recommendations.
  - Shipped AI-assisted CLI grouping, group descriptions, prompt batching fixes, and deterministic validation for LLM-assisted analysis.
- Catalyst Feature Parity & Platform Reliability
  - Delivered Catalyst RSPAN/Aurora 2.0 and StickyMAC feature-parity work across UI, config generation, NFO support, Scala grabbers, and TDL/Quark integration.
  - Improved customer-facing reliability across DHCP/ARP, CDP/LLDP, event logs, ping/traceroute, ACL/libyang, cable tests, and Summary Reports.
  - Authored reusable engineering docs for Config Analyzer planning, XPath authoring, mapping automation, extraction limitations, and test coverage evidence.

Software Development Engineer 2 — October 2023 - June 2024
Cleartrip by Flipkart — *Bengaluru, Karnataka*

- Platform Team
  - Orchestrated meticulous bug resolution strategies, ensuring seamless operation and enhanced reliability across systems.
  - Led the successful transition from Oracle to SQL databases, meticulously managing data transfer processes and ensuring seamless integration.
  - Spearheaded agile optimization campaigns for Ruby on Rails (RoR) codebase, harnessing advanced methodologies to bolster performance, scalability, and maintainability, amplifying user satisfaction and operational efficacy.

Software Development Engineer 2 — May 2022 - October 2023
1MG — *Gurgaon, Haryana*

- Subscriptions and care plan.
  - Enhanced the delivery and purchase experience of subscription plans, orchestrating seamless integration across multiple services to ensure a cohesive user journey and maximize revenue generation.
  - Upgraded a mission-critical web application from Rails 5 to the latest version, ensuring compatibility with modern frameworks and optimizing performance.

  - Developed CI/CD pipelines on Bitbucket for automated build, testing, and deployment, fostering seamless development workflows.

- E-Consults platform.
  - Mentored interns and freshers in Python and Ruby on Rails, sharing knowledge and helping develop new talent
  - Managed feedback and advertising for doctors, showcasing the ability to handle customer feedback and deliver exceptional results
  - Played a key role in developing the E-Consults platform, contributing to its success and enhancing the overall user experience for 1MG's customers

- Rule Engine
  - Strategized the implementation of a company-wide rule engine to process business rules.
  - Setup docker and Kubernetes pipeline for production-level deployment on the staging environment.

- Coupons and Offers
  - Parallelized the bulk coupon creation process reducing the coupon creation time by 90%
  - Maintained the service for bugs and fixes

Software Development Engineer 1 Sept 2019 - May 2022
1MG *Gurgaon, Haryana*

- E-Consults platform.
  - Designed and shipped slot-based e-consults, improving the overall user experience
  - Optimized queries and database performance to enhance the platform's speed and efficiency
  - Architected an E-consults flow experimentation framework for easy A/B experimentation and valuable business insights
  - Designed a widget service with a reusable architecture and backend reconfigurability, contributing to increased efficiency and flexibility
  - Implemented multiple security features in the API, including password policies, user throttling, SQL injection, query parameter sanitization, XRF, and script injection

- Built Customer Service Portal able to serve more than 10k requests per day.
  - Integrated API of different verticals to provide a robust frontend for ops.
  - Created a state preservation module for the front end to facilitate url sharing.
  - Designed a new image viewer plugin to occupy minimum space and reduce the code size in jQuery.

- Security
  - Advanced password policies to fortify user authentication mechanisms.
  - Effective user throttling techniques to mitigate potential brute force attacks.
  - Proactive measures against SQL injection vulnerabilities, including thorough query parameter sanitization.
  - Implementation of Cross-Site Request Forgery (XRF) protection to safeguard against unauthorized actions.
  - Deployment of safeguards against script injection attacks to ensure data integrity and user safety.

ACHIEVEMENTS AND CERTIFICATION

- Deep Learning Certification - Coursera (deeplearning.ai)
- Rank of 4281 in JEE ADVANCED 2015
- Rank of 5857 in JEE MAIN 2015

LEADERSHIP

- Secretary Science and Technology (IIT Ropar)